

THIBAUT FRANÇOIS

Post-doctoral Researcher · University of Surrey · Astrophysics

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RESEARCH INTERESTS

My research focuses on the **dynamics and mergers of massive black holes** across different mass scales — from **supermassive black holes** in the centers of elliptical galaxies to **intermediate-mass black holes** in dwarf galaxies, where off-center mergers present unique dynamical challenges. My approach combines high-resolution *N*-**body simulations**, cosmological simulations and theoretical studies of few-body dynamics.

EMPLOYMENT

2024–present **Post-doctoral researcher** — University of Surrey, UK
Working with Prof. Alessia Gualandris.

2021–2024 **PhD researcher in Astrophysics** — Observatoire Astronomique de Strasbourg, France
Working with Prof. Christian Boily and Dr Jonathan Freundlich.

EDUCATION

2021–2024 **PhD in Astrophysics** — Observatoire Astronomique de Strasbourg, France
Supervisors: Prof. Christian Boily and Dr Jonathan Freundlich.
Thesis: *Massive Black Holes in Dwarf Galaxies: Off-centre binary formation through gravitational capture.*

2019–2021 **Master's degree in Subatomic Physics & Cosmology** — Université Grenoble-Alpes, France
High honours.
Internship : *Phenomenology of primordial black holes at the center of galaxies.*
Master thesis: *Dark matter core formation from stellar and AGN feedback.*

2016–2019 **Bachelor's degree in Fundamental Physics** — Université de Montpellier, France
Highest honours.
Internship: *Quantum trajectories in curved space.*

TALKS & SEMINARS

MODEST 25 — 06/2025 | Talk
University of Surrey — 02/2025 | Seminar
GalPhases 24 — 08/2024 | Talk
MODEST 24 — 08/2024 | Talk
Observatoire Astronomique de Strasbourg — 10/2023 | Seminar
Royal Observatory Edinburgh — 09/2023 | Seminar
Conference of the French astronomical society (SF2A) — 07/2023 | Poster

TEACHING & COMMUNITY SERVICE

Teaching — 168h in total

- Practical thermodynamics and optics for biology undergraduate students (72h) — 2022, 2023.
- Maths for physicists (lectures & tutorials, 12h) — 2022.
- Practical fields & interactions for biology undergraduate students (60h) — 2021, 2022.
- Practical mechanics for physics undergraduate students (24h) — 2021.

Student supervision

Supervision of two physics undergraduate students for an intership on dark matter rotation curves, Observatoire de Strasbourg — 2023.

Community service

Local Organising Committee, National Cosmology and Galaxy Day — 2022, Strasbourg, France.
Created and organised the weekly PhD meetings at the Strasbourg Observatory.

GRANTS

Travel grant — International Astronomical Union — 2025

Young researcher travel funding — IRMIA++ — 2023

PhD studentship — University of Strasbourg — 2021–2024
Three years of research funding.

SKILLS

N-body simulations and dynamical modelling

Griffin, GyrFalcON, Bonsai, REBOUND, NEMO, Agama.

Cosmological simulations

IllustrisTNG, proficient in merger tree analysis.

Programming

Python, C++, bash, Mathematica, LaTeX.

Scientific computing

NumPy, SciPy, Pandas, Matplotlib, emcee, scikit-learn, Jupyter, SymPy.

Machine learning

PyTorch, TensorFlow/Keras.

HPC

High-performance computing, parallel computing (basic), SLURM.

Languages

French (native), English (fluent), Spanish (basic).

Additional training

FIDLE deep-learning training (30h): DNN, CNN, Embedding, RNN, Transformers, Autoencoder, VAE, GAN, Reinforcement.

Parallel computing training (21h).

Advanced cosmology course (50h), University of Jerusalem with Prof. Avishai Dekel.

SUMMER SCHOOLS

2022 **Group Theory** — International Colloquium on Group Theoretical Methods in Physics, Strasbourg, France
Lectures.

2019 **Particle and Astroparticle Physics** — LAPTh, Annecy, France
Summer school.

2018 **Physics Summer Camp** — CPPM/LAM, Marseille, France
Summer school.

PUBLICATIONS

– FIRST-AUTHOR PUBLICATIONS –

- 2026** **Shape, kinematics, and core properties of dry major merger remnants with supermassive black hole binaries**
François T. L., Gualandris A., Khonji N.
Submitted to *Monthly Notices of the Royal Astronomical Society*.
- 2026** **Scaling relations for binary black hole merger times from cosmological initial conditions**
François T. L., Gualandris A., Dehnen W.
Monthly Notices of the Royal Astronomical Society, 551(3).
- 2024** **Forming off-center massive black hole binaries in dwarf galaxies through Jacobi capture**
François T. L., Boily C. M., Freundlich J., Rozier S., Voggel K.
Astronomy & Astrophysics, 687, A203.

– CO-AUTHOR PUBLICATIONS –

- 2026** **Converging on the eccentricity of massive black hole binaries in Galactic mergers**
Gualandris A., Read J., Fastidio F., Chan J., François T. L., Khonji N., Dehnen W.
Monthly Notices of the Royal Astronomical Society, 546(1).
- 2025** **Dynamical traction and black hole orbital migration: I. Angular momentum transfer and a fragmentation-driven instability**
Boily C. M., François T. L., Freundlich J., Combes F., Melchior A.-L., Hénin Y.
Astronomy & Astrophysics, 701, A10.
- 2024** **Class Symbolic Regression: Gotta Fit 'Em All**
Tenachi W., Ibata R., François T. L., Diakogiannis F. I.
The Astrophysical Journal Letters, 969, L26.
- 2022** **The response of dark matter haloes to gas ejection: CuspCore II**
Li Z., Dekel, A., Mandelker N., Freundlich J., François T. L.
Monthly Notices of the Royal Astronomical Society, 518(4), 5356–5375.

SELECTED REFERENCES

Prof. Alessia Gualandris

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Dr Benoit Famaey

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